Unsupervised Domain Adaptation for Semantic Segmentation in Autonomous Driving

Project Group Details

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Abstract

Semantic segmentation is a critical perception task for autonomous driving systems, enabling pixel-level scene understanding of urban environments including road markings, vehicles, pedestrians, and traffic infrastructure. Deep learning-based segmentation models have achieved high accuracy on annotated datasets; however, they suffer from severe performance degradation when deployed on real-world data due to the *domain shift*—the discrepancy between synthetic training data and real-world conditions. This shift arises from differences in visual appearance, lighting conditions, texture characteristics, sensor modalities, and environmental variations.

Collecting and annotating large-scale real-world datasets for each deployment scenario is prohibitively expensive. Consequently, synthetic datasets such as GTA5 are leveraged for training; however, models trained exclusively on synthetic data exhibit poor generalization to real driving scenes. Unsupervised Domain Adaptation (UDA) addresses this challenge by transferring knowledge from labeled source domains (synthetic data) to unlabeled target domains (real data) without requiring additional manual annotations.

Recent advances in UDA for semantic segmentation employ diverse strategies, including self-training with pseudo-label refinement, adversarial feature alignment, instance-adaptive learning, extensive data augmentation, class-balanced learning to handle long-tail categories, and vision-language models to leverage semantic priors. Despite these advances, critical challenges persist: the generation and selection of high-quality pseudo-labels, mitigation of noisy labels during self-training, management of class imbalance particularly for rare objects, handling of unknown or out-of-distribution classes in target domains, and preservation of spatial consistency in segmentation boundaries.

This project investigates and implements state-of-the-art UDA techniques for semantic segmentation, specifically targeting the challenging GTA5-to-Cityscapes adaptation scenario. The methodology encompasses: (1) comprehensive literature survey of contemporary UDA approaches, (2) dataset preparation and exploratory analysis, (3) baseline model implementation

using standard architectures, (4) systematic implementation of selected UDA strategies addressing pseudo-label quality and class imbalance, and (5) rigorous performance evaluation using standard metrics (mIoU, per-class IoU) on real-world datasets. The project aims to develop a robust, generalizable segmentation model capable of effective performance on real-world driving scenes while establishing insights into the efficacy of different adaptation strategies. This work contributes to advancing practical autonomous perception systems by reducing annotation costs and improving cross-domain robustness.

Keywords: Unsupervised Domain Adaptation, Semantic Segmentation, Synthetic-to-Real Transfer, Autonomous Driving, Pseudo-Label Refinement, Computer Vision

Summary of Related Work (2024 Onwards)

Paper (Year)	**Problem Addressed**	**Key Contributions**	**Challenges Faced**
UDASS: Unified Domain Adaptive Semantic Segmentation (2025)	Unified handling of image and video UDA-SS; inconsistent treatment of spatial vs temporal domain shifts	Quad-directional Mixup (QuadMix) for intra- and inter-domain feature mixing; unified framework for image and video UDA-SS; optical-flow–guided spatio-temporal aggregation; SOTA performance on four UDA-SS benchmarks	Increased complexity from unified design; dependence on accurate optical flow for video; higher memory and compute cost due to feature mixup and aggregation

Paper (Year)	Problem Addressed	Key Contributions	Challenges Faced
UniMAP: Universal Domain Adaptation for Semantic Segmentation (2025)	Assumption of known shared label space; poor handling of private/unknown classes in target domain	Defines Universal DA for semantic segmentation (UniDA-SS); Domain-Specific Prototype-based Distinction (DSPD) with per-domain prototypes; Target-based Image Matching (TIM) to focus on common classes; introduces new UniDA-SS benchmark and outperforms BUS and other baselines	Requires careful prototype maintenance; sensitivity to pseudo-label quality for TIM; performance trade-offs in purely closed-set settings compared to specialized UDA methods
Rein: Stronger, Fewer & Superior (2024)	Limited generalization of CNN-based UDA; inefficient full-parameter tuning of Vision Foundation Models	Parameter-efficient fine-tuning of Vision Foundation Models for domain generalized semantic segmentation; instance-linked trainable tokens that refine features layer-wise; achieves strong performance on GTAV$\rightarrow$Cityscapes, ACDC, BDD100K, Mapillary using only synthetic or limited real data	Focuses on domain generalization rather than classical UDA; relies on large VFMs with high GPU memory; adapting token design to new backbones and datasets requires careful tuning

Paper (Year)	Problem Addressed	Key Contributions	Challenges Faced
Pixel-to-Pixel Contrast Learning (2024)	Pixel-level feature misalignment; pseudo-label noise degrading contrastive learning in UDA-SS	Combines pixel-to-pixel and pixel-to-prototype contrastive objectives; uses all target pixels with pseudo-label guidance; reports gains of about 5.1% mIoU on GTA5→Cityscapes and 4.6% on SYNTHIA→Cityscapes over DAFormer	Computationally expensive due to dense pixel-level contrast; prototype representations can become unstable under large domain shifts; improvements depend heavily on initial pseudo-label quality
BUS: Open-Set Domain Adaptation for Semantic Segmentation (2024)	Presence of unknown/OoD classes in target domain; misclassification of unknown regions as known classes	Proposes boundary-aware contrastive loss and shape-aware domain mixing; explicitly models unknown regions for open-set semantic segmentation; shows strong results compared to prior open-set UDA-SS methods	Performance degrades when unknown classes dominate the scene; requires careful thresholding/definition of unknown space; boundary estimation remains difficult in cluttered or low-resolution regions

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